

Task 8 — Descriptive Statistics and Technical Interpretation

Reading numbers honestly: what a summary statistic tells you, and what it quietly leaves out.

Siddeshwar — Week 2, Day 2 — 2 July 2026

Three measurements run through this note as a single thread, because descriptive statistics only become interesting when they're attached to something real. A mechanical rig is pulling five specimens of a new alloy to failure. A voltage regulator is being tested under low and high load. An enzyme assay is being run in triplicate, control against heated. Every idea below gets tested against one of these three before it's allowed to stand on its own.

1. Descriptive Statistics and Why They Matter

A descriptive statistic takes a column of numbers — five tensile-failure forces, six load-voltage readings, nine pH values — and returns a handful of numbers that stand in for the whole column: where it's centered, how far it spreads, whether it clusters or scatters. That's the entire job. It does not explain, it compresses.

The reason this comes before cleaning, visualization, feature engineering, or modelling is that every one of those later steps needs a working definition of "normal" for the variable in question, and that definition has to come from the data itself before you can act on it. You cannot decide the 340 N tensile reading is worth flagging until you already know the other four cluster near 118–124 N. You cannot pick a sensible bin width for a histogram of pH values until you know the range they span. Descriptive statistics are the baseline everything else is measured against.

What they will not do is tell you why the pattern exists. If the six voltage readings under high load show a standard deviation four times larger than the low-load readings, that number is a fact — but it says nothing about whether the cause is resistor tolerance, a loose connection, thermal drift in the regulator, or a cheap multimeter. The number points at a question; it does not answer it.

A dataset can be perfectly summarized and still poorly understood. Take the pH readings: mean 6.9, median 7.0, tight standard deviation — a summary that looks like a clean, well-behaved assay. Except three of those nine readings came from a batch where the buffer was accidentally swapped mid-run. The overall statistics are calm because the swap happened not to shift the numbers much this time, but the dataset is actually two different experiments quietly merged into one, and no amount of staring at the mean would reveal that. The number was innocent; the sampling wasn't.

2. Center and Spread

Mean is the sum of the values divided by how many there are. Median is the middle value once everything is sorted — the point where half the readings sit below and half above. Minimum and maximum are the extremes actually observed; range is the gap between them. Variance is the average squared distance of each value from the mean, and standard deviation is the square root of that — brought back into the original units so it's actually

interpretable (a variance in N^2 means little to a person holding a specimen; a standard deviation in N does). IQR, the interquartile range, is the width of the middle 50% of the data — the 75th percentile minus the 25th — deliberately blind to whatever sits in the tails.

Center answers "where does this data tend to sit." Spread answers "how much do individual readings disagree with each other." Two datasets can share an identical mean and still describe completely different experiments — one where every reading lands within a hair of the average, one where the average is a compromise between wildly different values.

The mean turns misleading exactly where the tensile test shows it: four specimens fail near 120 N, a fifth fails at 340 N because it was accidentally cut from a different stock. The mean climbs to about 165 N — a number that describes no specimen in the batch and overstates the alloy's real strength. The median, sitting at 122 N, shrugs the outlier off almost entirely, because it only cares about position in the sorted order, not magnitude.

A high standard deviation in an experimental measurement is a claim about the process, not just the numbers. Six voltage readings under a fixed high load with a large standard deviation say the setup itself is unstable — a marginal connection, an unregulated supply, or a load that isn't actually constant — and that instability needs chasing down before the mean voltage means anything as a spec number.

IQR earns its keep exactly when the mean and standard deviation start lying — skewed data, or data with a genuine outlier sitting in the tensile set. Reporting "IQR of the four typical specimens is about 4 N" tells a materials engineer something the full-range standard deviation, dragged around by the 340 N result, cannot: how consistent the alloy actually is once the anomaly is set aside for separate investigation.

3. Distribution Shape and Sample Size

A distribution is the full shape a numeric variable takes on — not just its center and spread, but whether it's symmetric or lopsided, whether values bunch into separate clumps, whether there are gaps where no reading ever falls. Two variables can share a mean and standard deviation and still have completely different distributions: one smooth and bell-shaped, one with two distinct humps.

Skewness says the tail on one side is longer than the other — the pH readings might skew low if contamination events, when they happen, drag values sharply downward rather than upward. Clusters can reveal that two different conditions were folded into one dataset — the control and heated enzyme samples, plotted together, will show two separate humps rather than one, because they are not the same population. Gaps often trace back to instrument resolution or a threshold effect, such as a current sensor that cannot register anything below its noise floor.

Five tensile-failure readings and five hundred are not the same kind of evidence, even if they produce an identical mean. With five, one unusual specimen can move the mean by tens of newtons and there's no way to tell if it's a fluke or a real second population; with five hundred, a single outlier is diluted almost to nothing, and the shape of the distribution starts to reflect the material rather than the luck of which five specimens got tested. Trusting a mean, a variance, or a visual pattern requires knowing how many readings built it — a smooth-looking histogram from five points is not evidence of a smooth process, it's an accident of too little data to show the roughness.

The clearest small-sample trap here is the tensile batch itself: three specimens tested, two near 120 N and one at 300 N, produces a mean around 180 N that could easily be reported as "this alloy is notably strong." With three data points there is no way to know whether the 300 N reading represents a genuine strength characteristic or a mislabeled sample from a different batch — the sample is too small to distinguish a real signal from a single bad reading.

4. Outliers and Domain Context

An outlier is a reading that sits far enough from the rest of the values — for the same variable, the same instrument, the same experimental condition — that it doesn't look like it came from the same process as its neighbors. That definition is relative, not absolute; "far" only means something once you know how tightly the rest of the readings cluster.

An outlier is not automatically a mistake. Real processes occasionally produce genuine extreme events — a specimen that really was cut from a stronger region of stock, a transient current draw that really did spike during motor start-up, a cell culture that really did have an unusually fast reaction. Deleting every statistical outlier on sight risks deleting the most scientifically interesting row in the dataset.

The right first move is almost always investigation, not deletion — checking the instrument log, the experimental notes, the raw sensor trace — and removal is only justified once there's a specific, documented reason (a known sensor fault, a transcription error, a confirmed contamination event) rather than the fact of being numerically extreme.

- Biochemistry: a pH reading of 3.1 against a cluster sitting near 7.0 could mean the buffer was contaminated, the wrong stock solution was drawn, or the electrode wasn't calibrated — but it could also be a real acid-producing reaction worth a second look before it's written off.
- Electronics: a current spike to roughly 400 mA against a steady 50 mA baseline could be inrush current at switch-on, a genuine transient overload, or an early sign of a developing short — each implies a different next step, so the spike needs a timestamp and a cause before it's dismissed as noise.
- Mechanical: a force reading of 340 N against neighbors near 120 N could be sensor saturation, a units mix-up (kgf logged as N), or a specimen that is genuinely, rarely stronger than the rest of the batch — worth re-testing before it's assumed to be an error.

5. Plot Selection and Interpretation

A histogram shows how one numeric variable is distributed across its full range — how many of the nine pH readings fall in each narrow band from 3 to 9 — and is the right first plot whenever the question is "what does this variable's spread actually look like."

A box plot summarizes center, spread, and outliers for one or more groups at a glance — comparing the voltage spread under low load against high load side by side, with any extreme readings marked individually rather than hidden inside the whiskers.

A bar chart compares counts or totals across categories — the number of specimens tested from material A against material B — and should never be reached for when the variable being shown is actually continuous, which is a common misuse.

A scatter plot shows the relationship between two numeric variables measured on the same observations — tensile force plotted against displacement for each specimen, to see whether the material's failure behavior looks linear, brittle, or gradual.

A line plot is for a variable tracked across an ordered sequence, almost always time — voltage recorded every millisecond during a load-switching transient, where the order of points carries information a scatter plot would throw away.

Every one of these plots needs a title, labeled axes, and units, because a plot without them is exactly as ambiguous as a bare number without a unit: a curve labeled only "5" on the y-axis could be volts, milliamps, or newtons, and the reader has no way to tell which claim is being made.

6. Correlation, Causation, and Confounding

Correlation measures how consistently two variables move together — as one goes up, does the other tend to go up (or down) alongside it. Causation is the stronger claim that a change in one variable is what produces the change in the other. Two variables can rise together for reasons that have nothing to do with either one driving the other: coincidence over a small sample, or — more commonly — a third variable driving both at once.

That third variable is a confounder. In the running example, ambient lab temperature could rise over the course of an afternoon, and both the enzyme reaction rate and the voltage regulator's output could drift upward alongside it — not because faster reactions cause higher voltage, or the other way around, but because the room got warmer and pushed both readings in the same direction. Reporting "reaction rate correlates with voltage" without checking temperature would hand someone a causal story that doesn't exist.

The same trap shows up outside the lab: attendance and exam score typically move together, but concluding attendance causes the score ignores that a third factor — a student's underlying motivation or interest in the subject — plausibly drives both showing up to class and studying for the exam.

7. Domain Measurement Interpretation and Comparability

Averaging a pH reading, a voltage, a current, and a force together produces a number, but not a meaningful one — the four variables are not measuring the same physical property, do not share a unit, and do not even share a dimension. The result is arithmetic performed on quantities that were never comparable in the first place, and it cannot be mapped back onto anything a domain scientist would recognize.

A unit is not decoration attached after the fact — it is part of what the number means. "120" is not a force until "N" is attached to it, and swapping N for kgf silently changes what the number claims by a factor of roughly 9.8, without the digit itself giving any warning that this happened.

Before any comparison, values need to be grouped by domain, by measurement type, by unit, and by the condition they were recorded under — because a comparison is only valid once every one of those four things matches on both sides.

Three valid comparisons

- Mean regulator output voltage under low load versus high load, same circuit, same unit, same test rig.
- Mean pH of the control samples versus the heated samples, same assay, same electrode, same unit.
- Mean force at failure for material A specimens versus material B specimens, same tensile rig, same unit, same test speed.

Three invalid or misleading comparisons

- Averaging pH values together with voltage readings, since neither shares a unit or a physical dimension with the other.
- Comparing a force dataset recorded in newtons against another recorded in kgf without converting first, since the raw digits differ by close to a factor of ten for no physical reason.
- Pooling current readings from two circuits built around different resistor tolerances or rated at different supply voltages and treating the merged set as one homogeneous population.

Before analyzing data handed over by another team, the missing pieces worth asking for are: the exact unit and scale for every numeric column, the calibration status of the instrument that produced it, the experimental condition or protocol each batch of readings was collected under, and whether any cleaning, filtering, or exclusion has already been applied before the file reached you.

8. Final Reflection

A statistic can pass every arithmetic check and still mislead, because arithmetic doesn't know what the numbers mean. A mean computed correctly across pH and voltage values is still meaningless once it leaves the calculator, because the two were never the same kind of quantity. A standard deviation reported without its unit tells you nothing about whether the spread is trivial or alarming. Five readings can produce a razor-tight variance that looks convincing purely because there wasn't enough data for the real roughness of the process to show up — and the same five readings, if one happens to be a genuine outlier, can swing a mean into describing a specimen that doesn't exist. Two variables correlating tightly can look like a causal story right up until a confounding variable — temperature, motivation, time of day — turns out to be driving both. None of these failures show up as an error in the calculation; the formula was applied correctly every time. What was missing was context the formula was never designed to hold: what was measured, in what unit, under what condition, from how many samples, and whether something else in the room might have been quietly pulling the strings.

Review Questions

1. If mean and median are far apart, what does that suggest about the data?

It suggests the data is skewed, or contains at least one extreme value strong enough to pull the mean away from where most of the readings actually sit — the tensile batch with its 340 N outlier is exactly this case.

2. Why is standard deviation not enough without knowing the unit and context?

A standard deviation of "8" is uninterpretable on its own — 8 V on a 5 V regulator is alarming, 8 mV is not, and without knowing the experimental condition it was measured under, there's no way to judge whether that spread is normal process noise or a sign something is wrong.

3. Is every outlier wrong? Give one case where an outlier is valuable.

No. A specimen that genuinely fails at 340 N because it came from a stronger region of stock is a real, useful data point — it's the kind of reading that might point toward a better manufacturing process rather than something to be deleted.

4. Why is 105 percent attendance a valid data type but an invalid value?

It's still a number, so the data type (numerical, percentage) is correct — but attendance is bounded between 0 and 100 by definition, so a value outside that range signals a recording or calculation error rather than a type problem.

5. Why is an average of pH, voltage, current, and force meaningless?

Because the four variables measure different physical quantities with different units and different natural scales; averaging them produces a number that isn't tied to any of the four properties it was computed from.

6. Which is more robust to outliers: mean or median? Why?

The median, because it only depends on position in the sorted order rather than magnitude — one extreme value can shift it by at most one rank, while the same value can move the mean by an arbitrarily large amount.

7. What does a box plot show that a bar chart does not?

A box plot shows the spread, median, and individual outliers of a continuous variable; a bar chart shows totals or counts across categories and has no concept of spread or outliers at all.

8. If attendance and score have positive correlation, what can you conclude and what can you not conclude?

You can conclude the two tend to rise and fall together across the students measured. You cannot conclude that attending more classes causes the higher score, since a confounder such as motivation could be driving both.

9. What is one example of a statistically correct statement that is scientifically useless?

"The average of all pH, voltage, and force readings collected this week is 41.7" — the arithmetic is correct and the number is completely disconnected from anything a domain scientist could act on.

10. What extra metadata would you ask for before analyzing Biochemistry, Electronics, or Mechanical measurements?

The unit and scale of every numeric column, the calibration state of the instrument used, the exact experimental condition or protocol each reading was taken under, and whether the data has already been filtered, cleaned, or had any exclusions applied before it was handed over.